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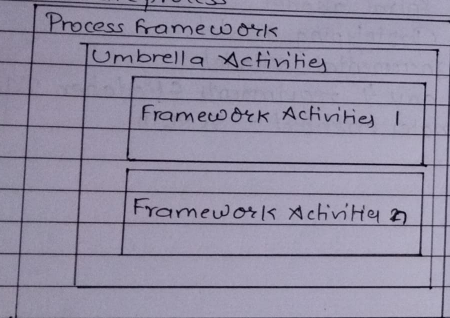
* Answer:-

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Q] Explain in brief the Software Process framework.

- A software engineering process is the model selected for managing the creation of software from customer initiation to the release of the finished product.

- Software process



Software framework
common to
multiple product
not
unique
to
one
project
not
specific
to
one
project

- Framework activities Include #1

Work tasks
Work products
QA checkpoints
Project milestones

A generic process framework following activities it composes :-

1) Requirement/Information gathering :

Communication with client to understand project requirements for defining software features and then the document is made called Software

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requirement specification (SRS).

2) Design : ^{like DRD}

Diagrams or Plan to be designed which defines the workflow that is to follow. and is made/considered w.r.t. SRS.doc. to know what to add/remove.

3) Implementation :

An longest phase of all phases where code generation is carried out which has a proper developed code or result as per the requirements followed.

4) Testing :

Developed code is to be tested here in this phase to check whether all the requirements of client are fulfilled or not by software and if they are corrected and modified s/w code.

5) Deployment :

Delivers software to Client and collect feedback from clients based on evaluation software support

6) Maintenance :

Once the s/w is deployed actual problems from user's perspective will come up. and it's team's responsibility to solve user's prob. time to time and can add/delete additional features as per user's requirements.

- Umbrella Activities Include :-

- 1) Software project tracking & control
- 2) Risk Management
- 3) Software Quality assurance (SQA) → to ensure integrity of s/w.
- 4) Software configuration management (SCM)
- 5) Formal technical review (FTR) →

SQA | SCM
FTR

Q7 mention any 4 requirement Elicitation method.

- Requirements Elicitation is the practice of researching & discovering the requirements of systems for users, customers and other stakeholders.
- There are different ways to identify customer reqs

1) Interviews :

Organizⁿ may take var. kinds of interviews as they are imp way for gathering requirements. Support to cover any missing requirements as no. of people participate in this process.

2) Survey :

Collecting req. is economically beneficial bcz it collects reqs. from a large no. of persons at a same time. Most useful for capturing clear factual info.

3) Domain Analysis :

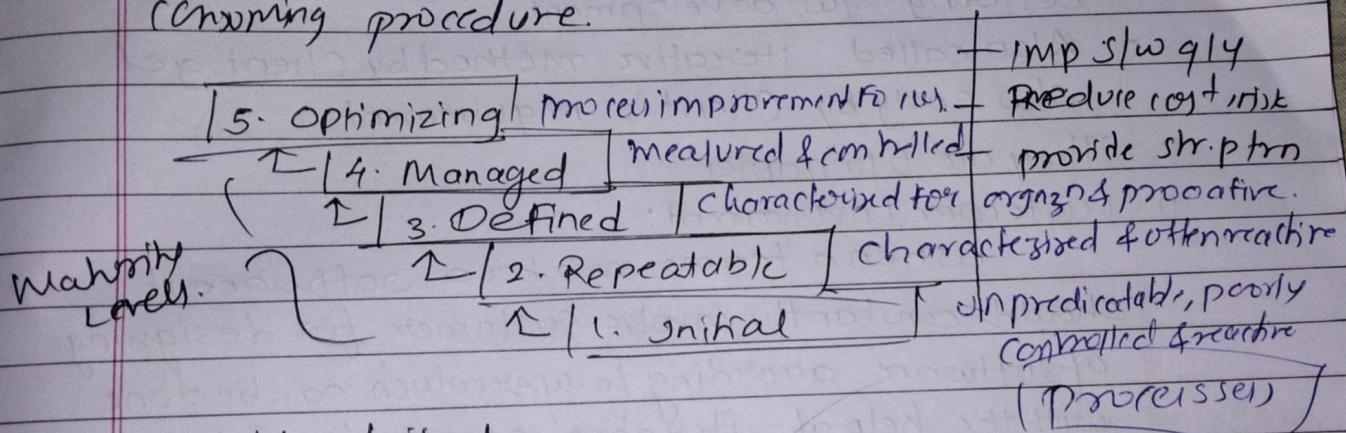
Each s/w put into some domain category. The experienced persons in that domain can be a great support to study general and specific reqs.

4) Brainstorming :

An informal debate held betⁿ different stake holders and all their suggestions documented for further reqs. analysis.

Q1 → "Capability Maturity Model (CMM)"

- The CMM invented by Software Engineering Institute (SEI) states a growing sequence of levels of a soft. development orgnⁿ.
- CMM is a mechanism used to create and redefine s/w creatn procedure of orgnⁿ.
- Higher the level, better the s/w development proced. so reaching every level is a cost consuming & time consuming procedure.



1) Level 1 - Initial

s/w development process is unpredictable w.r.t. to time & cost. things depend on current staff, if staff changes, process changes.

2) Level 2 - Repeatable

Planning & management of new project is based on experience of similar projects.

3) Level 3 - Defined.

Training prog. are implemented so that staff members have skills reqd.

level 4 - Managed.
process are measured & controlled focus on
q/ty. management.

level 5 - Optimizing
if there is prblm in s/w then this level's
approach is to not get the same prblm in future.

→ 9] Agile Methodology

- It values the small changeable increment for bettering s/w development.
- Also called iterative method by Client get s/w early by Client's requirement.
- They are 5 principles

1) Customer Involvement:

when a team makes a software it should constantly involve customer for designing of software according to user which can be done with the help of iterative approach. prototype user.

2) Incremental delivery:

By iterative approach prototype has been given to user for feedback & again a better prototype is made, update will go one by one.

3) People not process:

Software designer can predevelop these software acc to process & their own logic.

4) Embrace Change:

Change can happen acc. to user & adapt the changes. So it could be flexible.

5) Simplicity:

Allow simplicity to build s/w.

Q] SCRUM

→ • SCRUM

- One of the most popular Agile methodology.
- Scrum is a lightweight, iterative & incremental framework.
- Scrum breakdowns the development phase into stages or cycles called "sprints".
- It is used in s/w for managing & controlling s/w development.

• Character.

1. Stakeholder :

person want to build s/w buy to company for making s/w.

2. Product owner :

Task list for s/w what to add/not.

3. Scrum Master :

help to convey the task to developer team.

4. Development team :

Implement the req^r. code to project to build the s/w.

5. Customer :

End users) the person who uses the s/w.

dedicated thereby managing
Scrum team has scrum master and product
constant communications on the daily basis.
Key words: Backlog, sprint, daily scrum, scrum
owner

Advantages:

- Freedom & Adaption
- High quality, low risk product
- Reduce the development time up to 40%.
- Scrum customer satisfaction is very important

- Reviewing the current sprint before moving to new one

Disadvantage:

- More efficient for small team size
- No changes in the sprint.

SCRUM:

1. Stakeholders
2. Product
3. Scrum Master
4. Development Team
5. Customers.

Stakeholder visits a company which makes software
as stakeholder gives their demand to the company for a
particular software which they want.

A software team is selected for this task. In this team a
product owner is there which will create a product Backlog.
product owner mentions all the task required for making
a software in the Backlog. And these task have a priority
ie the task which is having more priority are at the top
of the list & the least important at bottom

Scrum Master makes the communication b/w the task of the
Backlog to the development team through scrum meetings

In scrum meeting ^{have} a software developer, product owner & ~~scrum~~ scrum master
all are present.

These scrum meeting are also known as sprint planning meetings.

In these sprint planning meeting the things which are discussed
as considering time to decide which top priority task can
be completed from Backlog which is known as sprint Backlog.

A sprint Backlog consists of current task which the
team is currently going to complete.

This meeting will held everyday. It is a standup meeting
duration of 15-20 mins. In this meeting 3 questions are
asked which are 1. what the team has done b/w the
previous meeting ~~until~~ to this meeting. 2. what problems are
faced 3. How to complete or solve problem.

Then development team makes software to the customer
& gives feedback. It is a iterative model.

④ NFR

system.

fast security, logical, reliable,
portability.

Types

① Product req.

execution in specific way deliver

② Organiza req.

policy rules.

③. External req.

so/w communicate with others/w.